Discussion paper for CIE review of the eastern Bering Sea walleye pollock stock assessment

May 2025, CIE review

Jim Ianelli, Carey McGilliard, Sofia Wassermann

2025-05-01 00:00

Executive Summary

The work presented below was based on the same data sets used in the 2024 assessment. The first task was to update the model runs to include the 2023 data. The model runs were updated and the results are presented below. The model runs were then used to compare the FT-NIR data with the traditional data sources (BTS and ATS). The results of these comparisons are presented below. The model fits to the data are presented in the tables at the end of the document. The first section steers towards the R library used for processing model results for EBS pollock. The second section presents the comparisons of the FT-NIR data with the traditional data sources. The third section compares results of different software platforms (e.g., SS3, CEA, and AMAK) for the assessment. The fourth section shows results comparing newer model-based estimates from survey data.

R Library for assessment compilations (ebswp)

The `ebswp` library is a collection of functions for compiling and analyzing results from the stock assessment models for the eastern Bering Sea walleye pollock (Ianelli (2025)). The library is designed to work with the ADMB code and depends on a consistent format for the report files generated. The link in the reference provides access to some notes on applications as vignettes to the package (e.g., . here).

Comparing FT-NIR compositions with traditional microscope age estimations

The fish age and growth lab at the AFSC have endeavored to expand on new technologies (Helser et al. (2019)).

FT-NIRS data consistency comparisons

Examining the abundance-at-age (for design-based estimates of the bottom-trawl age compositions) we show that in general, the TMA estimates show a high level of consistency by cohort (Figure 1). For these same data, when applied to the FTNIRs age-estimation method, the consistency was lower (Figure 2). The extent that this is a concern is unknown.

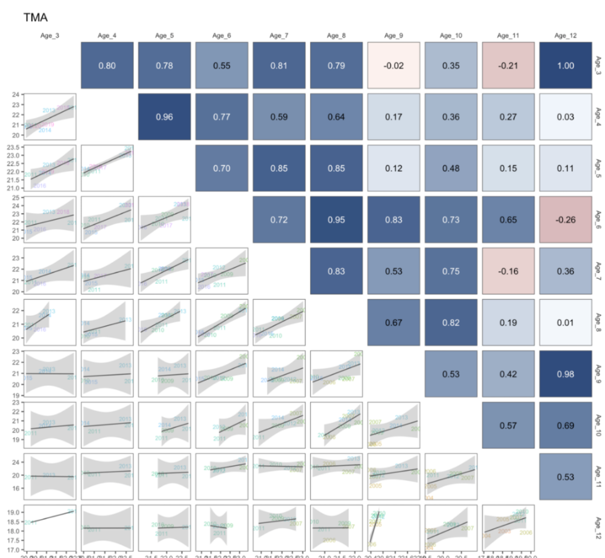

Figure 1: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for traditional microscope age estimates. The cohorts are shown in the labels.

When preparing the data for the assessment, we plotted out the proportions-at-age for the TMA data and noted the differences in the estimates using the FTNIRs age estimates (Figure 3,

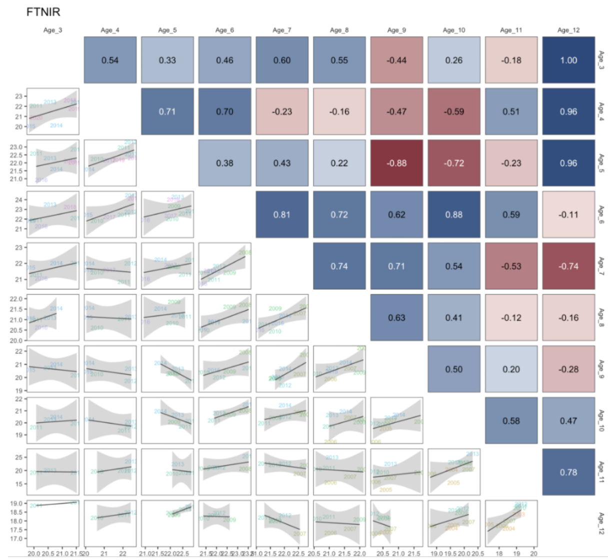

Figure 2: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for FTNIRs age estimates. The cohorts are shown in the labels.

Figure 4, and Figure 5). Other preparations involved estimating the age-determination errors for each method following that of Punt et al. (2008). For the FTNIRs data, we optionally estimated separate age-estimation error matrices by gear type. The appearance of the fleet-aggregated and fleet-specific input data files is shown in Figure 6.

For the assessment model, the alternatives are shown based on the coverage of data and availability among types of ageing is shown in Figure 7. These figures are intended to show what data are available for the model testing and the different ways of treating age determination error matrices.

Comparisons of fit to FT-NIRS models

The layout of model runs comparing the traditional microscope ages (TMA) with the FT-NIRS ages is shown in Table 1. The models are run with the FT-NIRS data with different age-error matrix specficiations. The model fits to the data are shown in Table 2. The table shows the goodness-of-fit

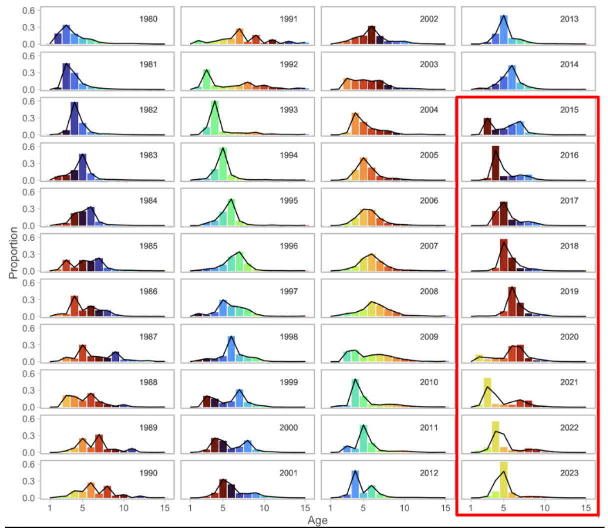

Figure 3: Proportions at age for recent years for the fishery data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

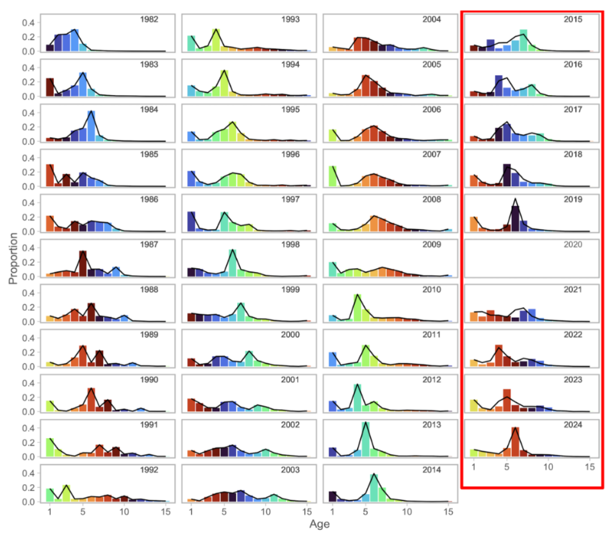

Figure 4: Proportions at age for recent years for the bottom-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

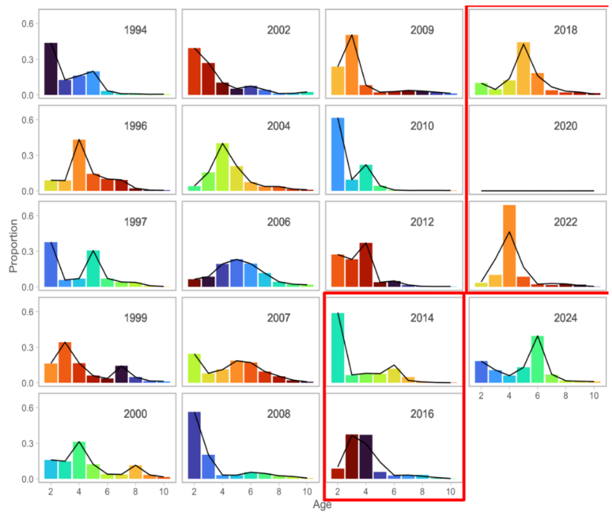

Figure 5: Proportions at age for recent years for the acoustic-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

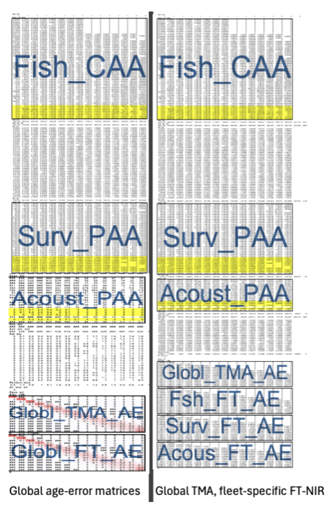

Figure 6: Schematic of input datafile showing the breakout of the fleet-specific FTNIRs age-error matrices (right panel) compared to that where the global FTNIRs age-error matrices are used (left panel).

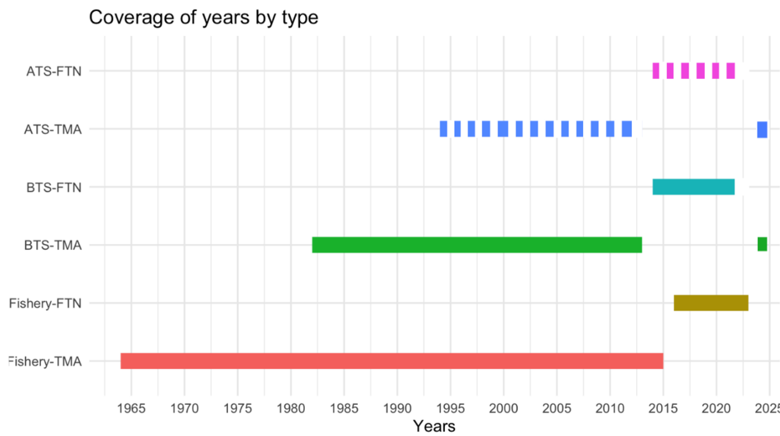

Figure 7: Data extent by age composition for the models with FTNIRs age estimates. The data types are for the Fishery, bottom trawl survey (BTS), and acoustic trawl survey (ATS) and the years of coverage are shown in the x-axis.

Table 1: Configuration of different model options testing the impact of different treatment of the age composition data and ageing error.

	Model description	
Name	Age type	Age-error
Base	TMA	Global TMA
FT-NIR 1	TMA + FT-NIR	Global TMA, global FT-NIR
FT-NIR 2	TMA + FT-NIR	Global TMA, disaggregated FT-NIR

Table 2: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	Base	FT-NIR fleet-specific	FT-NIR fleets aggregated
RMSE BTS	0.150	0.152	0.151
RMSE ATS	0.167	0.158	0.163
RMSE AVO	0.216	0.215	0.217
RMSE CPUE	0.097	0.097	0.097
SDNR BTS	0.890	0.880	0.900
SDNR ATS	0.930	0.880	0.910
SDNR AVO	0.950	0.940	0.960
Eff. N Fishery	357.670	381.050	336.190
Eff. N BTS	159.430	152.910	147.940
Eff. N ATS	233.170	232.690	220.510
Catch NLL	2.960	2.630	3.590
BTS NLL	23.910	23.380	24.430
ATS NLL	7.750	6.940	7.470
ATS Age1 NLL	6.930	8.850	8.170
AVO NLL	8.240	8.030	8.320
Fish Age NLL	416.810	613.630	558.740
BTS Age NLL	286.120	312.980	329.560
ATS Age NLL	42.550	41.060	41.990
NLL selectivity	212.700	215.450	215.820
NLL Priors	20.330	20.470	20.380
Data NLL	807.120	1031.620	995.390
Total NLL	1110.900	1332.810	1298.930

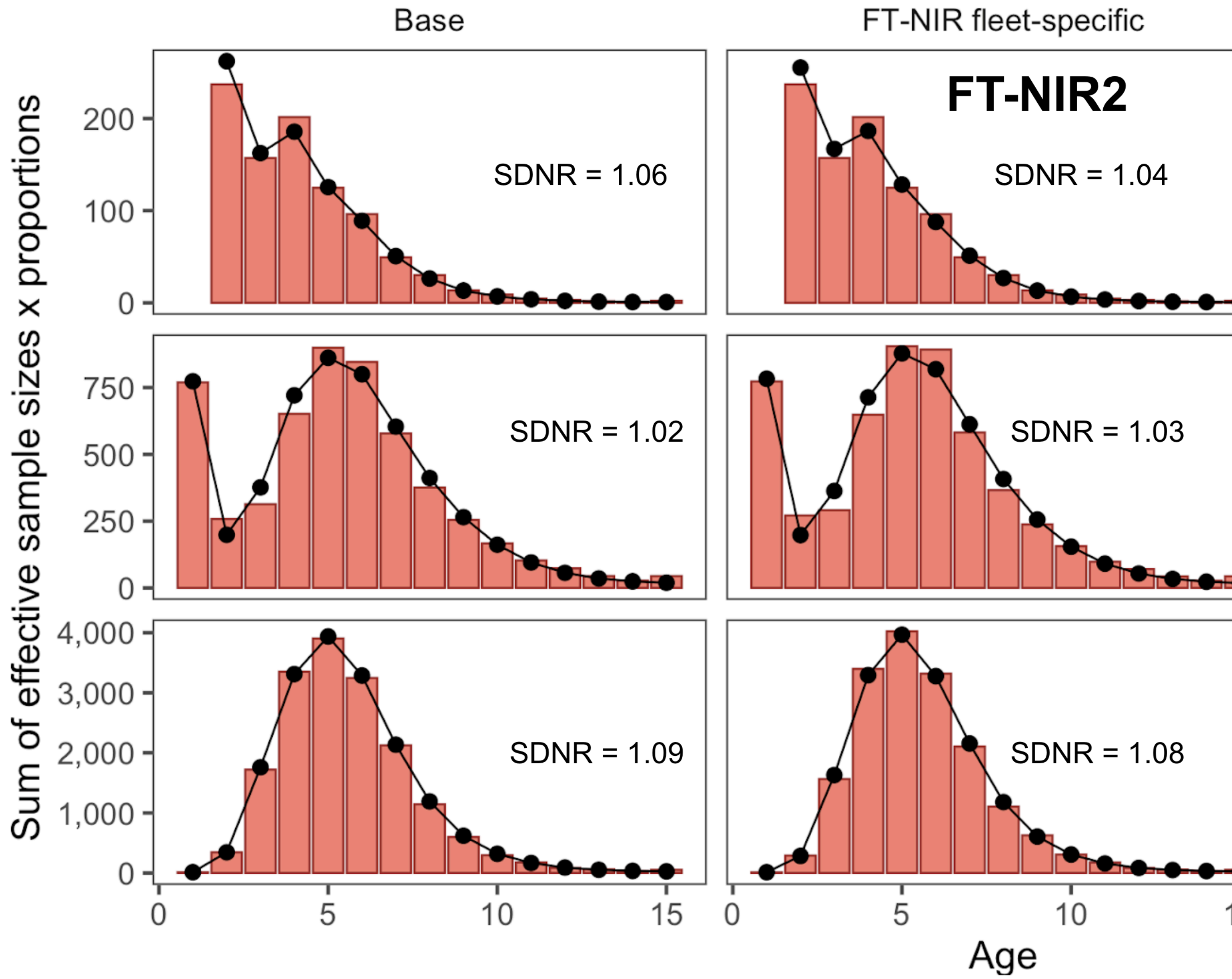

Figure 8: One-step ahead residuals for different model specifications for the fishery composition data

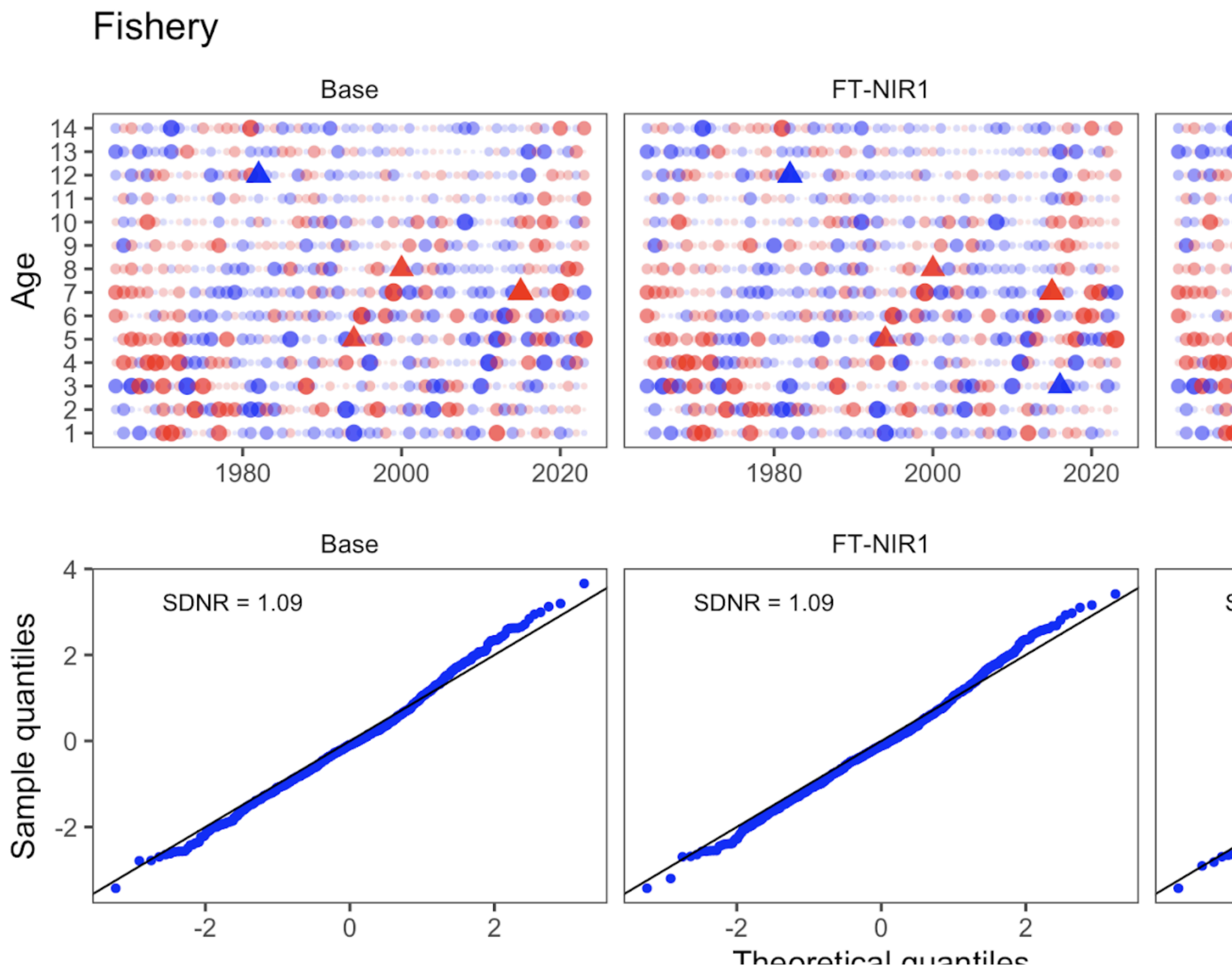

Figure 9: One-step ahead residuals for different model specifications for the fishery composition data

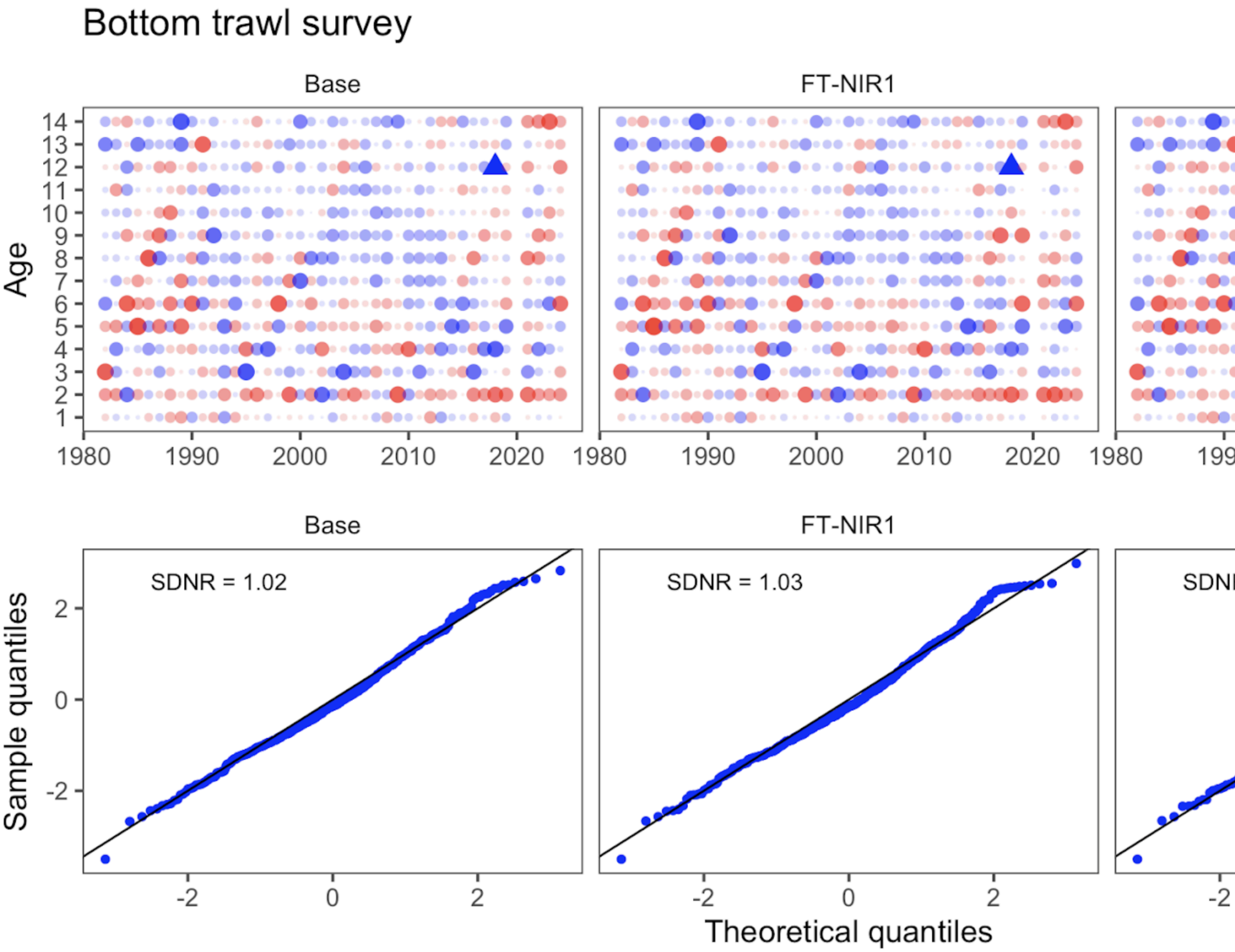

Figure 10: One-step ahead residuals for different model specifications for the bottom-trawl survey composition data

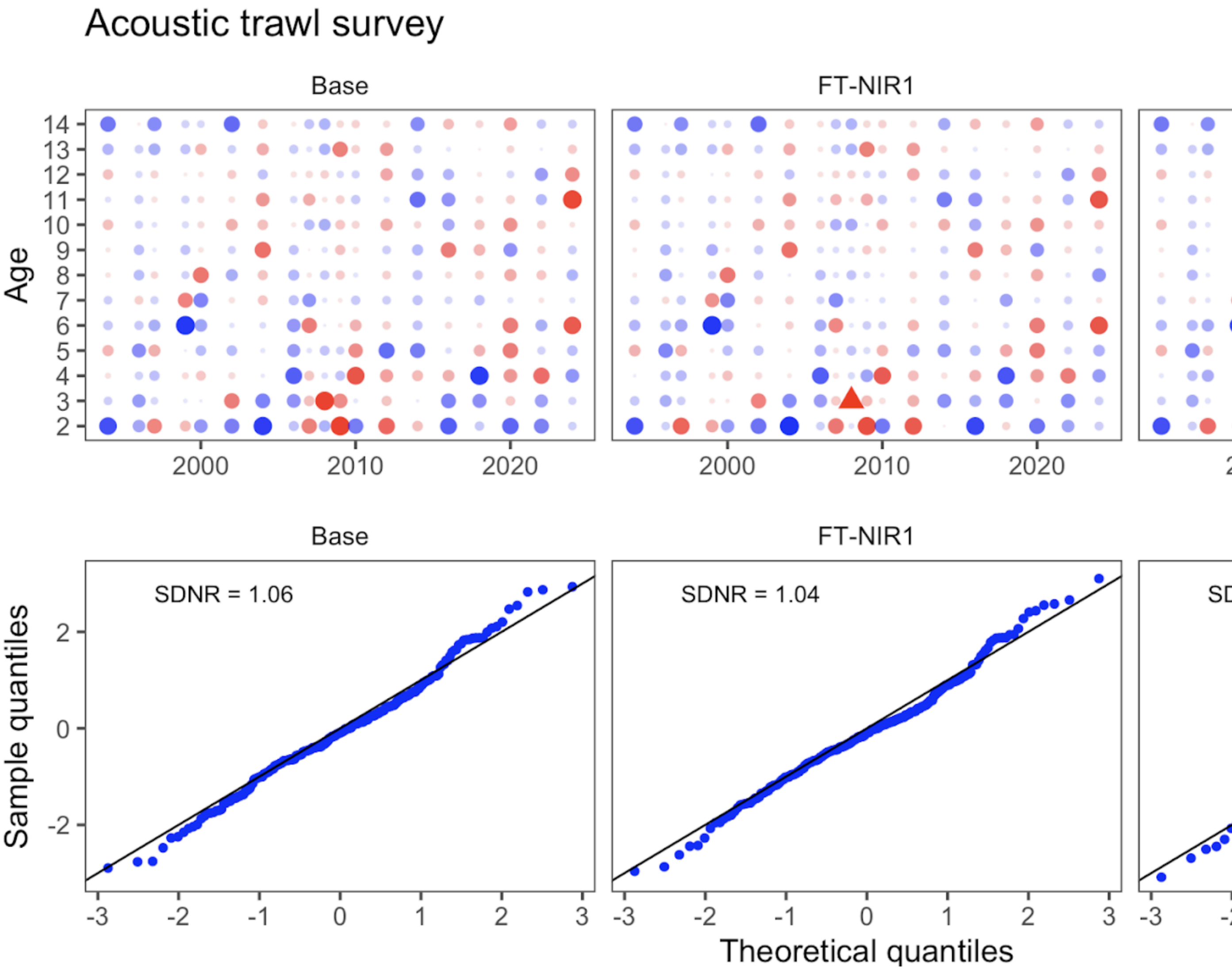

Figure 11: One-step ahead residuals for different model specifications for the acoustic-trawl survey composition data

Leave-one-out sensitivity analysis

Here we present model runs that downweight one source of age composition data at a time, as well as a run that downweights all survey biomass indices. Table 1 shows the combinations of data types included for this analysis.

Results show that the model is sensitive to the inclusion of the … data (Table 2)

Table 1: Configuration of different model options testing the impact of different data components

	Model description					
Dataset	Drop all BTS	Drop all ATS	Drop all indices	Drop AVO	Drop BTS Age composition	Drop ATS Ag
BTS age 2+ Index	X		X			
ATS age 2+ Index		X	X			
ATS age 1 Index		X	X			
AVO Index			X			
BTS comps	X				X	
ATS comps		X				X

Table 2: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	2024 accepted	Drop BTS	Drop ATS	Drop AVO	Drop indices	Drop BTS age
RMSE BTS	0.157	0.152	0.157	0.157	0.157	0.152
RMSE ATS	0.180	0.178	0.188	0.191	0.187	0.178
RMSE AVO	0.229	0.231	0.238	0.241	0.238	0.231
RMSE CPUE	0.093	0.092	0.093	0.093	0.098	0.092
SDNR BTS	0.950	0.850	0.950	0.960	0.960	0.850
SDNR ATS	1.000	0.980	0.010	1.050	1.040	0.980
SDNR AVO	0.980	0.980	1.010	1.080	1.050	0.980
Eff. N Fishery	1,182	1,344	1,190	1,185	1,176	1,344
Eff. N BTS	248	84	254	249	245	84
Eff. N ATS	269	228	164	269	273	228
Catch NLL	3	2	2	3	3	2
BTS NLL	31	22	31	31	31	22
ATS NLL	9	8	0	10	10	8
ATS Age1 NLL	11	10	0	11	11	10
AVO NLL	9	9	9	11	10	9
Fish Age NLL	167	125	166	167	165	125
BTS Age NLL	194	426	192	194	195	426
ATS Age NLL	30	30	57	30	29	30
NLL selectivity	196	167	189	195	195	167
NLL Priors	20	20	20	20	19	20
Data NLL	465	260	418	457	416	260
Total NLL	717.520	484.630	661.740	708.010	674.170	484.630

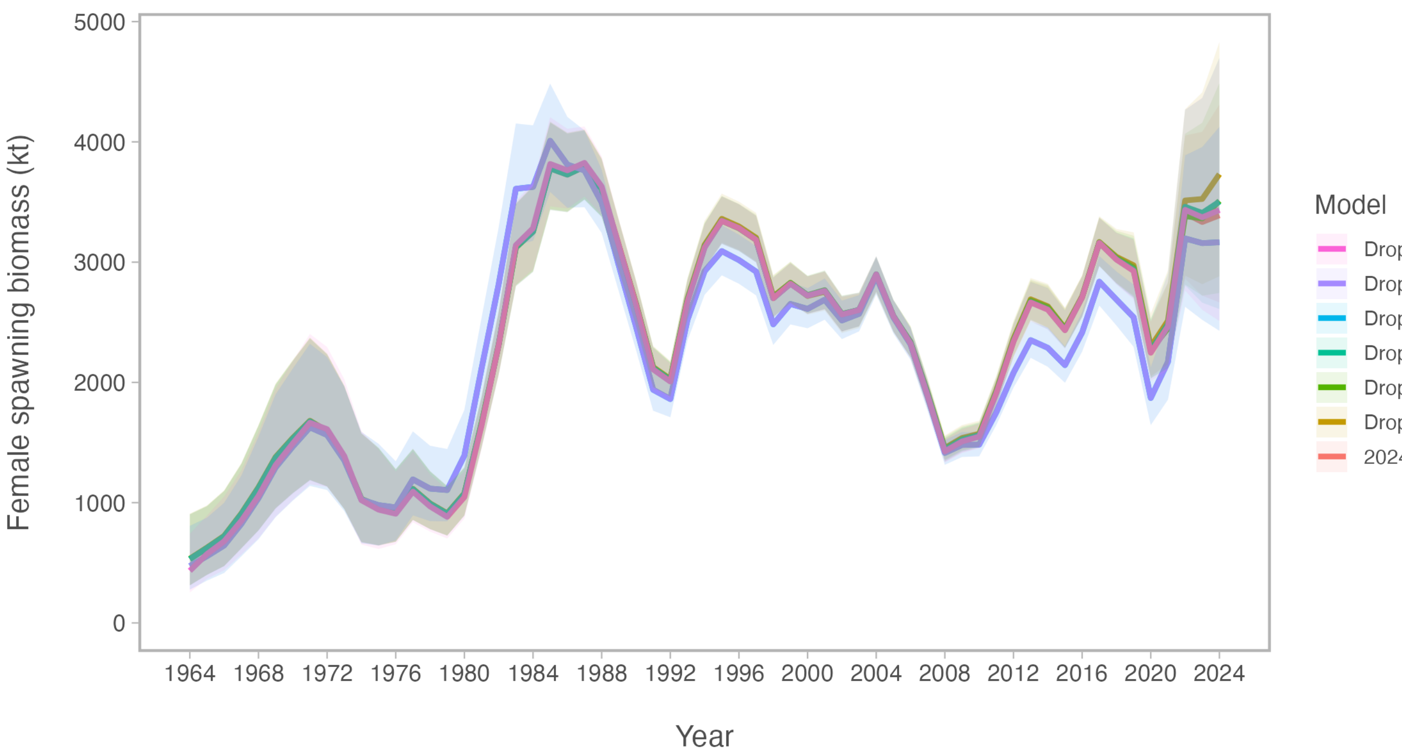

Figure 12: Model results comparing models that downweight one data source at a time

Alternative assessment software

Stock-synthesis 3

Stock synthesis 3 (SS3) is a well establish framework and is common in many diverse settings. (Methot and Wetzel (2013)) For EBS pollock, we extended an initial framework based on the configurations used for Pacific hake (Grandin et al. (2024)) since there are many similarities in the type of data available. Namely, empirical weight-at-age, acoustic survey data, and indices of 1-year olds. Figure 13 shows the model results comparing the last year's model some alternative model platforms. Comparing the SS3 output with three different configurations on selectivity constraint shows xxx (**?@fig-ss3ts**). Comparing the SS3 output with last year's reference model shows that xxx (**?@fig-ss3tsref**).

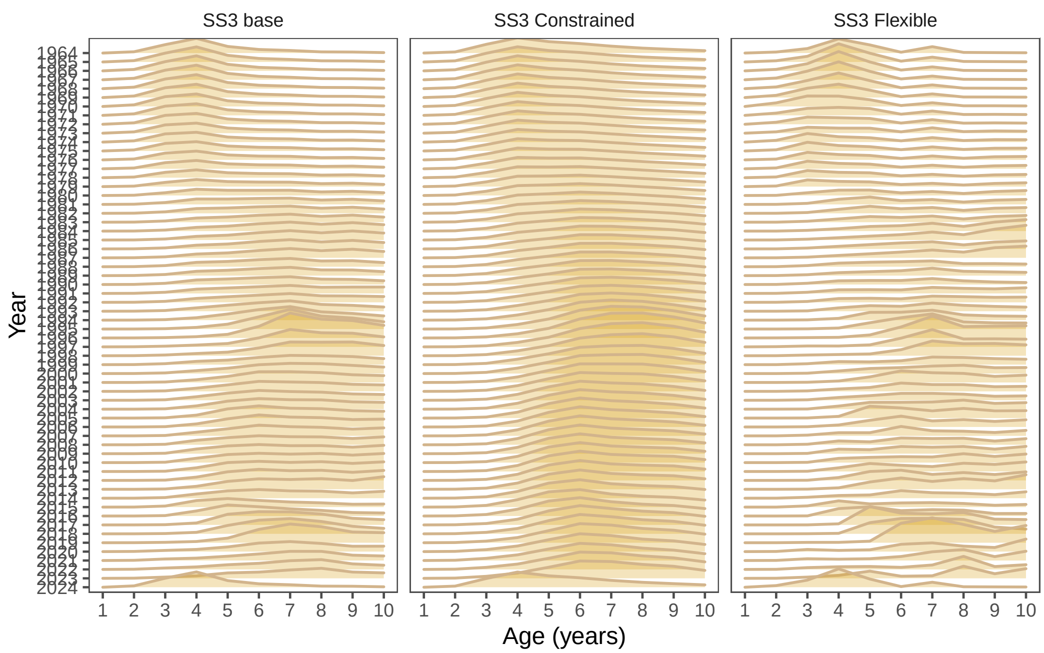

Figure 13: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

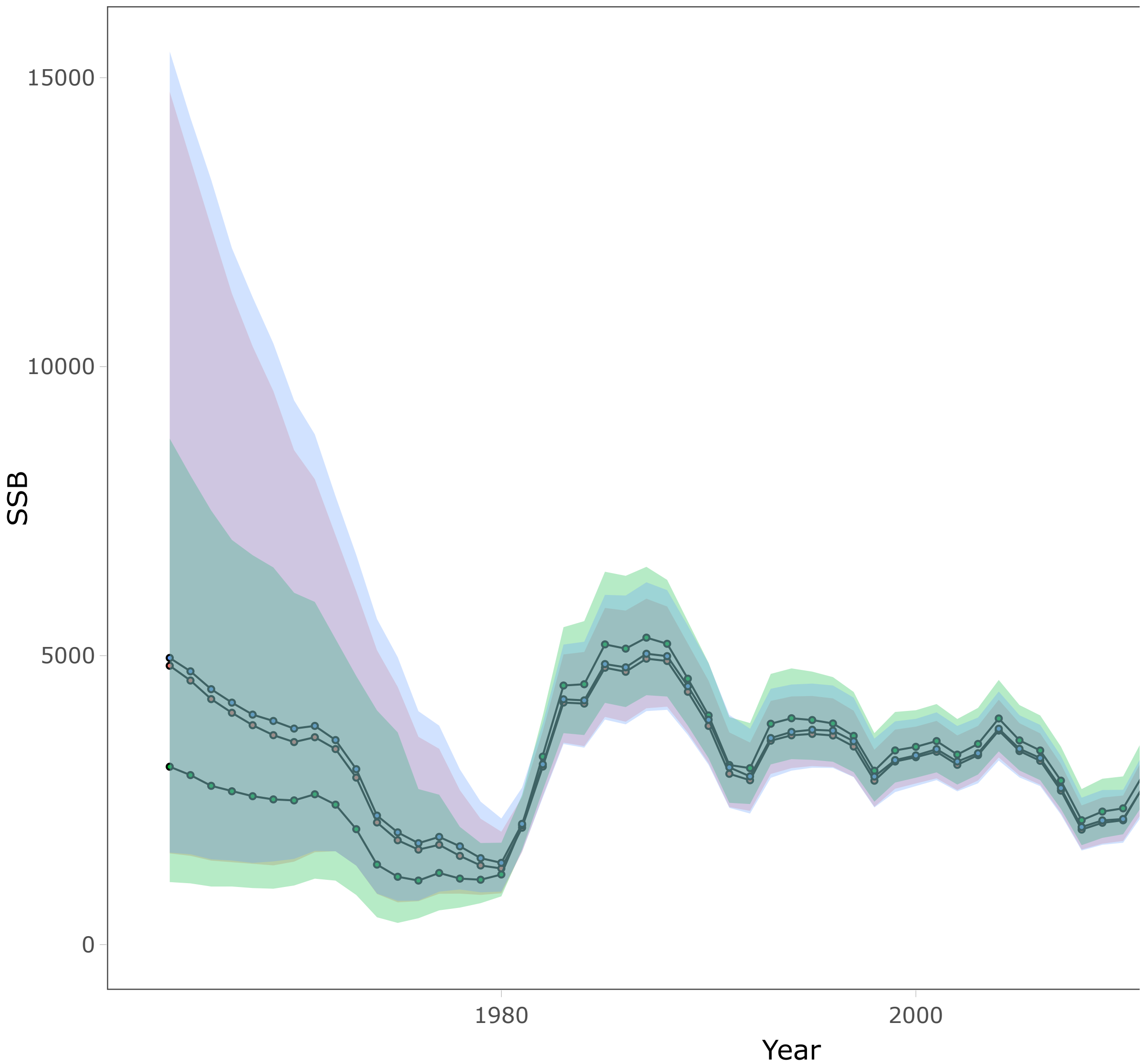

Figure 14: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

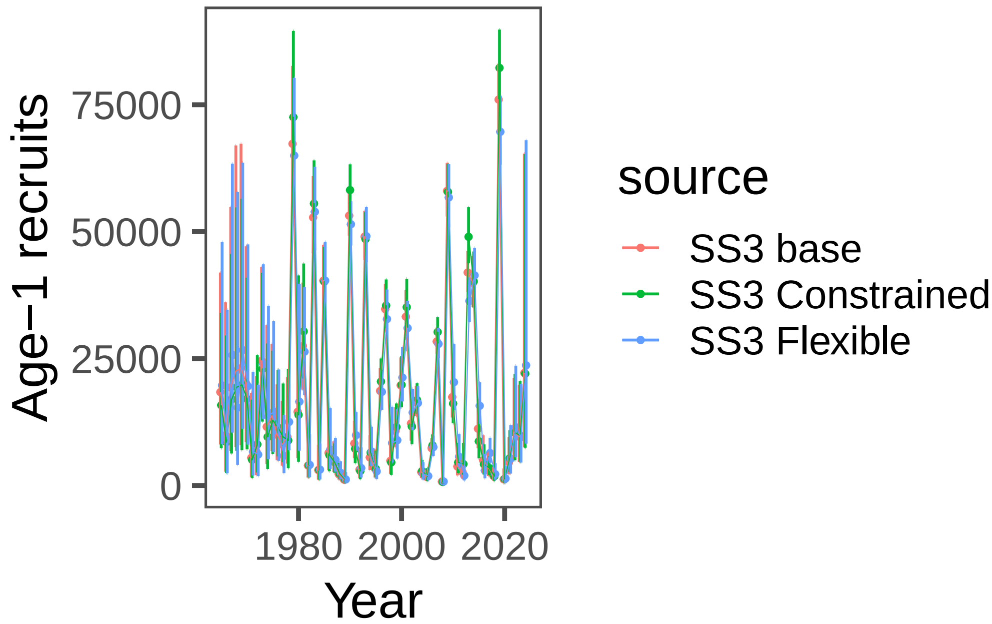

Figure 15: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

SAM

The Stock Assessment Model (SAM) is a state-space model written in TMB (Nielsen and Berg (2014), Nielsen et al. (2021)). We attempted to configure the settings to have similar levels of flexibility to the reference model used for catch advice in 2024 and the settings and results can be found here). Comparing "selectivity" in the SAM model to the reference model, we see broad similarities but slightly lower inter-annual variability in the SAM run (Figure 18). The spawning biomass trend largely overlaps in the uncertainty (**?@fig-sam_ts**).

WHAM

The "Woods Hole Assessment model" (WHAM) is written in TMB and is a fully state-space model (Stock and Miller (2021)).

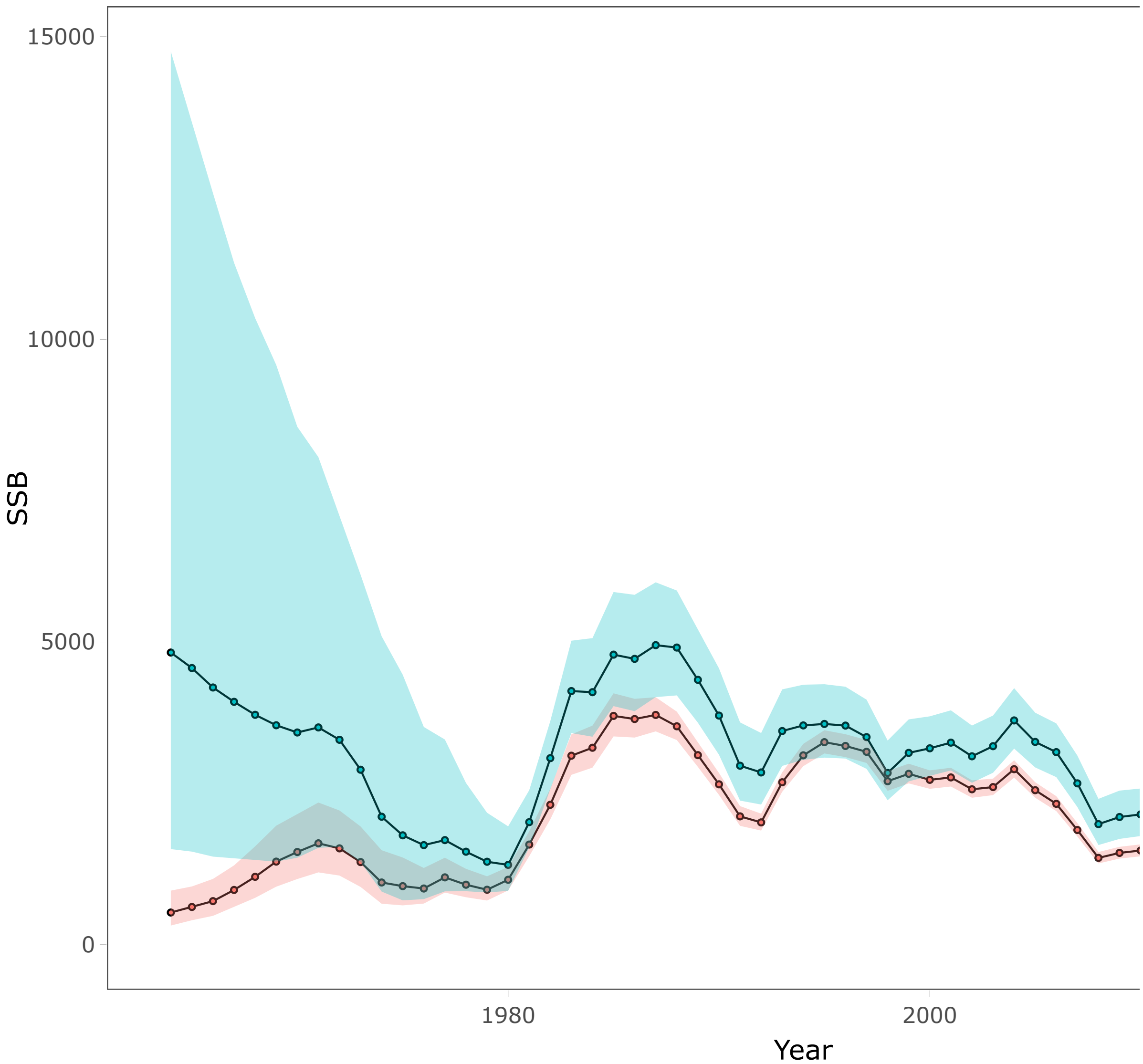

Figure 16: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

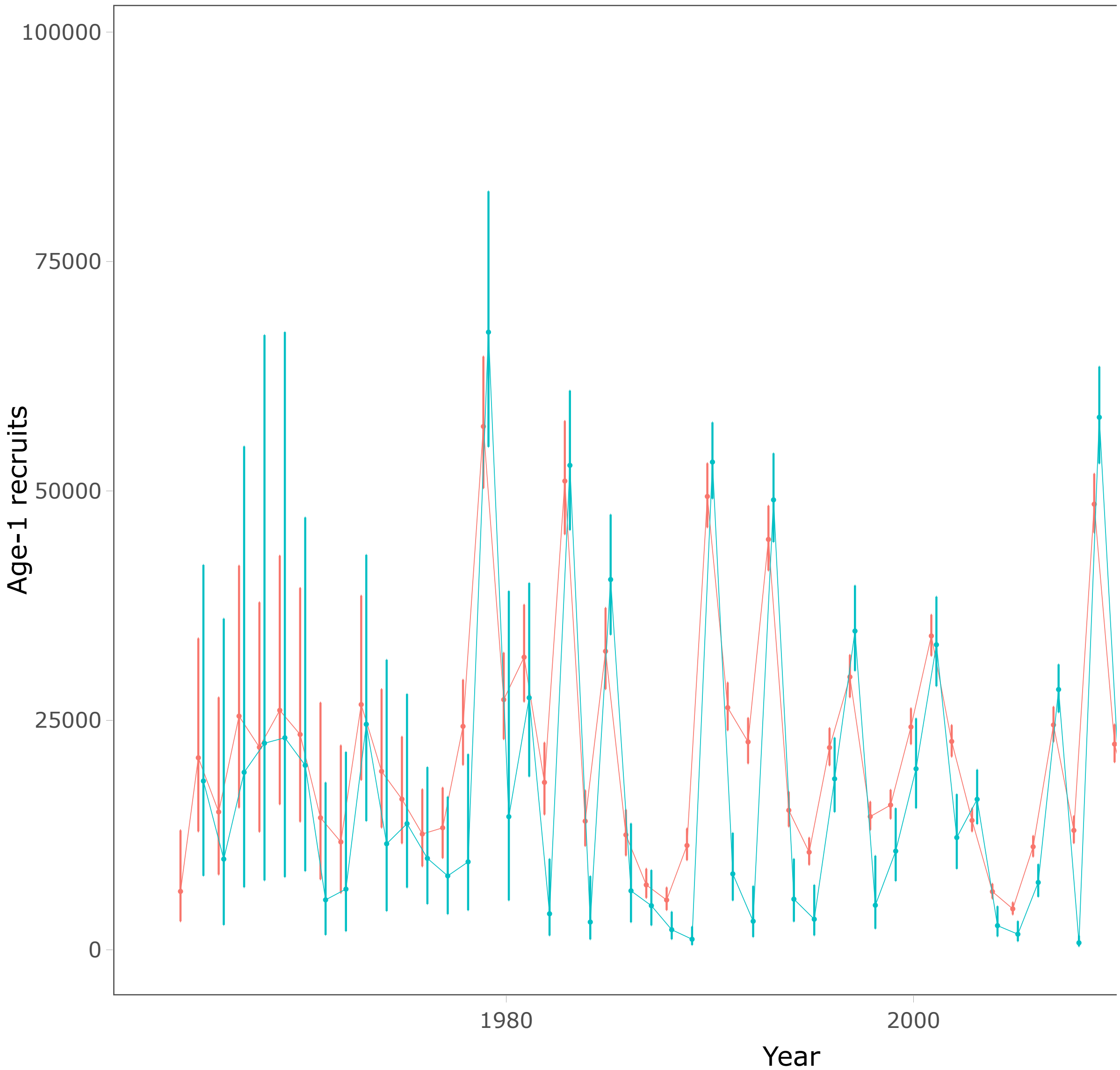

Figure 17: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

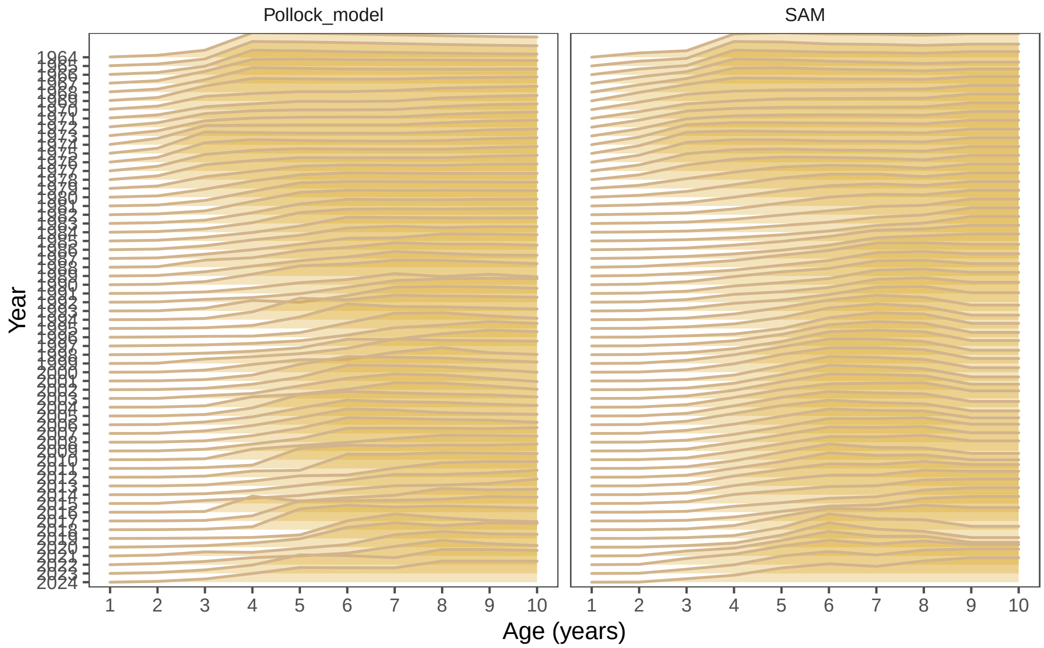

Figure 18: Model results comparing last year's model with the SAM model.

RTMB model

Work at AFSC has developed a general model which has flexibility to specify regions and a variety of options including random effects. While this model is not yet fully developed, we present preliminary results here. The model is written in TMB and is similar to the WHAM model. We pursue this model to explore the potential for a more flexible model that can be used for more explicit considerations of stocks linked to different regions (e.g., the US and Russian portions of the EBS). Also, this model to add sex-specificity and random effects (M. L. H. Cheng et al. (2025)).

CEATTLE model

The Rceattle model is developed by the AFSC and is written in TMB (Adams et al. (2022)). The model is an extension of the ADMB version used for multi-species model in the Eastern Bering Sea (Jurado-Molina, Livingston, and Ianelli (2005), Holsman et al. (2024)) and has a number of newer features. It is generalizable to include a variety of data types using lengths and can be set up for species tracked for sexual dimorphic growth and other processes. Since it has been recast into TMB, it allows for the ability to specify random effects in a number of processes. One development of the model is to have the flexibility to bridge among extant single-species modeling platforms by using it in "single-species" mode. To illustrate that, we

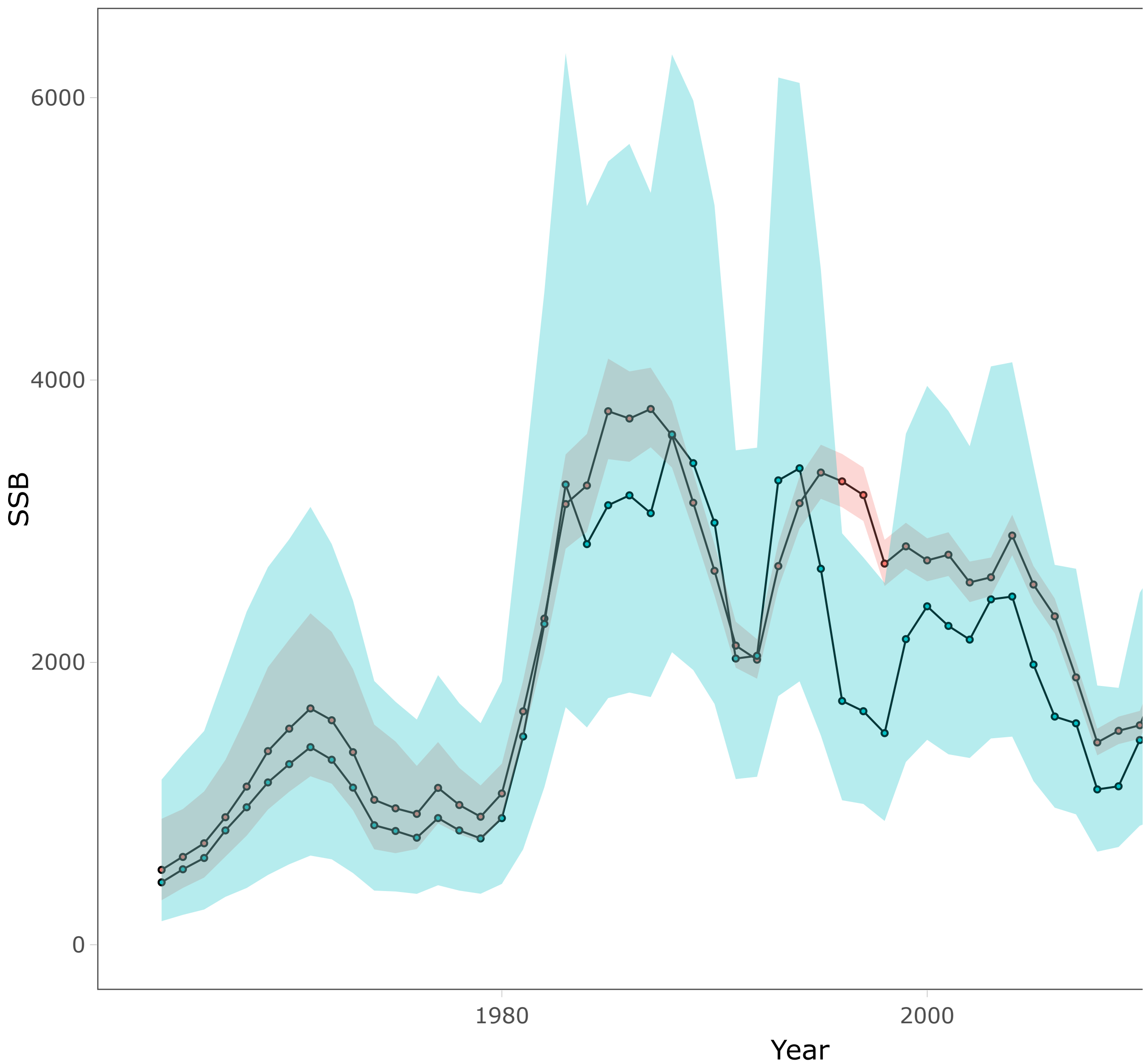

Figure 19: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

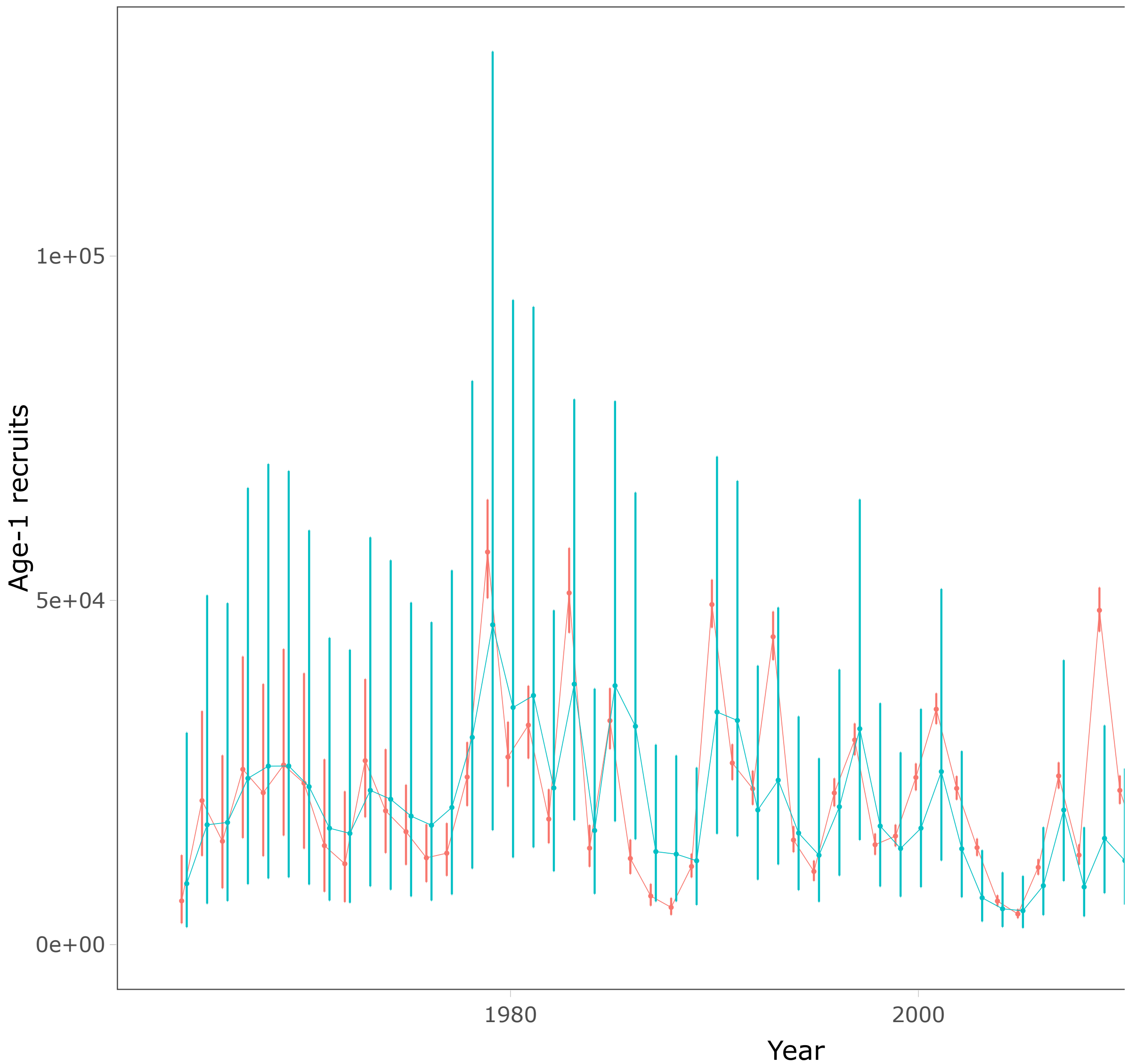

Figure 20: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

introduce an application of Rceattle to the EBS pollock data set. The model is not yet fully developed, but we present preliminary results for consideration.

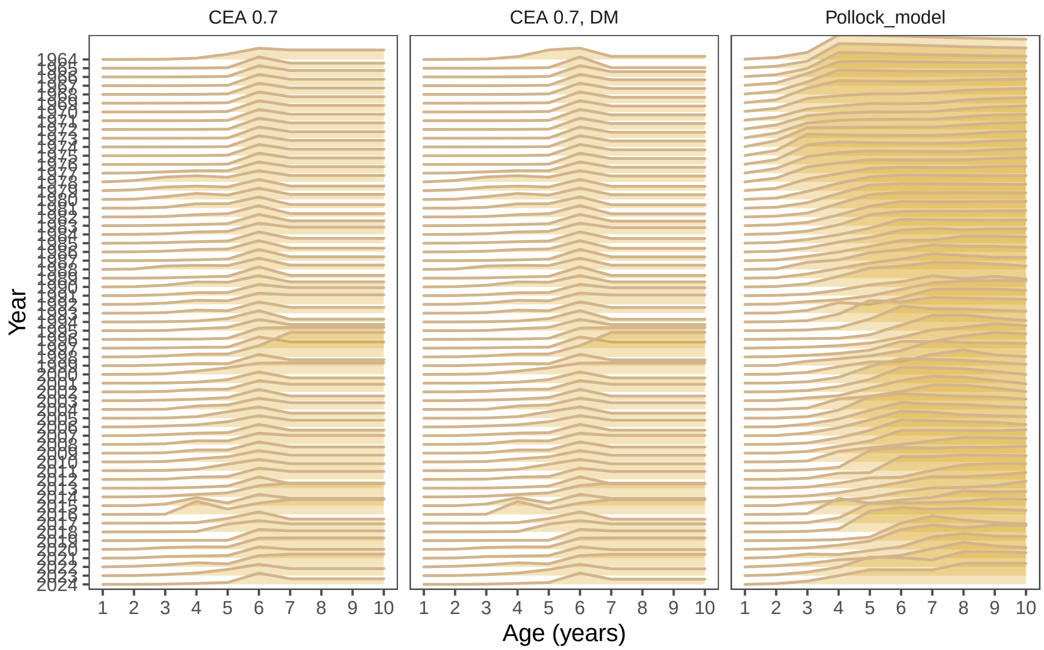

Figure 21: Model results comparing last year's model with the Rceattle model.

Spatial modeling of the EBS pollock stock

In development is a spatially disaggregated general model by colleagues at the Auke Bay lab and the University of Alaska, Fairbanks (M. Cheng (2025)). Figure 26 shows the model results comparing time-varying selectivity whereas **?@fig-spock_ts** shows the model results time series. The difference in the age-1 recruits is likely due to the fact that the spatial model currently has constant M-at-age compared to the base model.

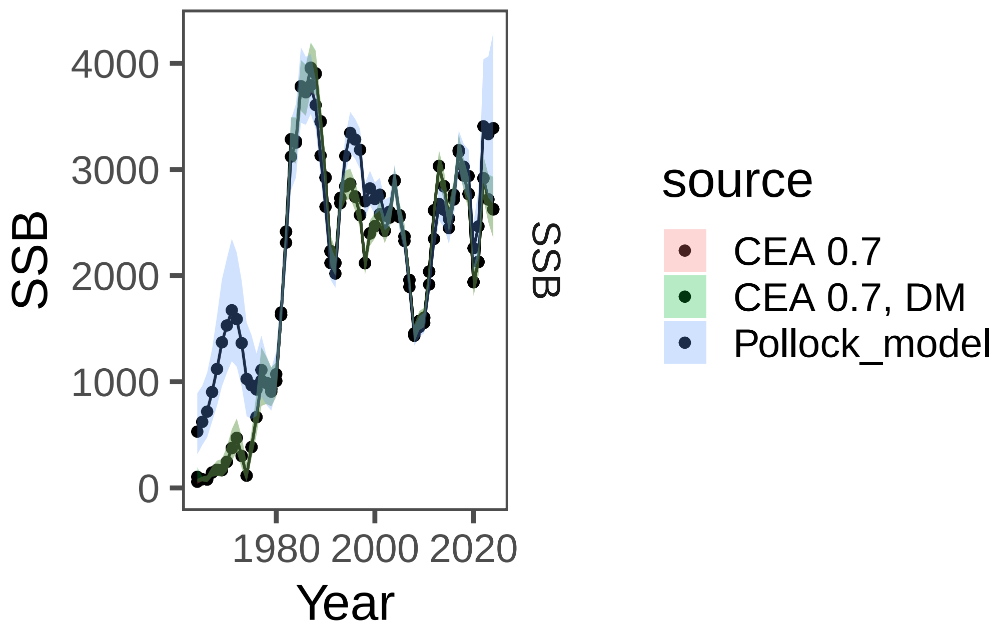

Figure 22: Model results comparing last year's model with the Rceattle model.

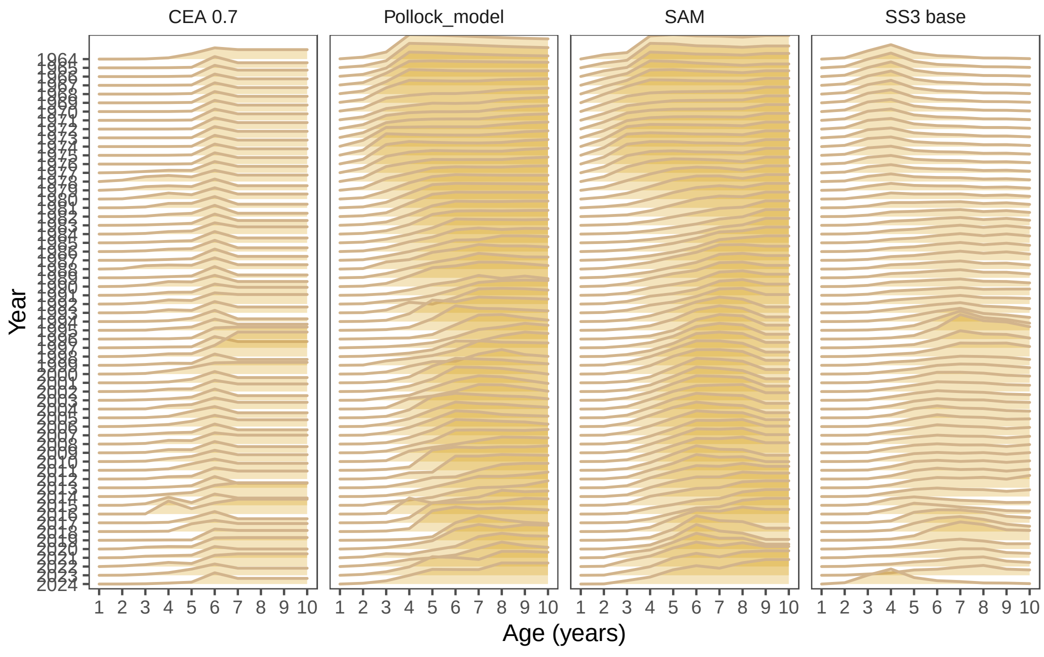

Figure 23: Model results comparing last year's model with the Rceattle model.

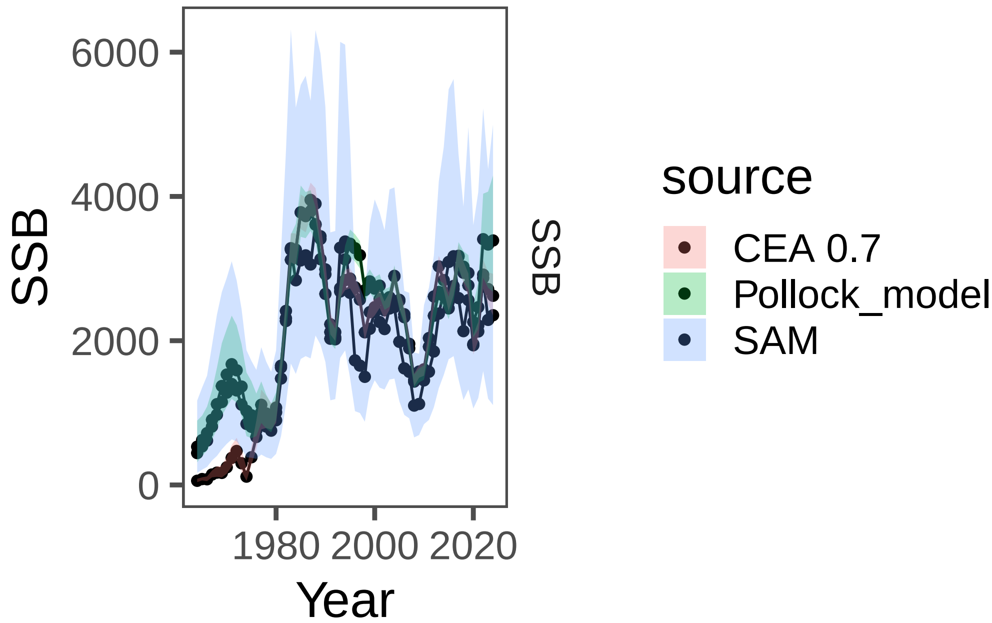

Figure 24: Model results comparing last year's model with the Rceattle model.

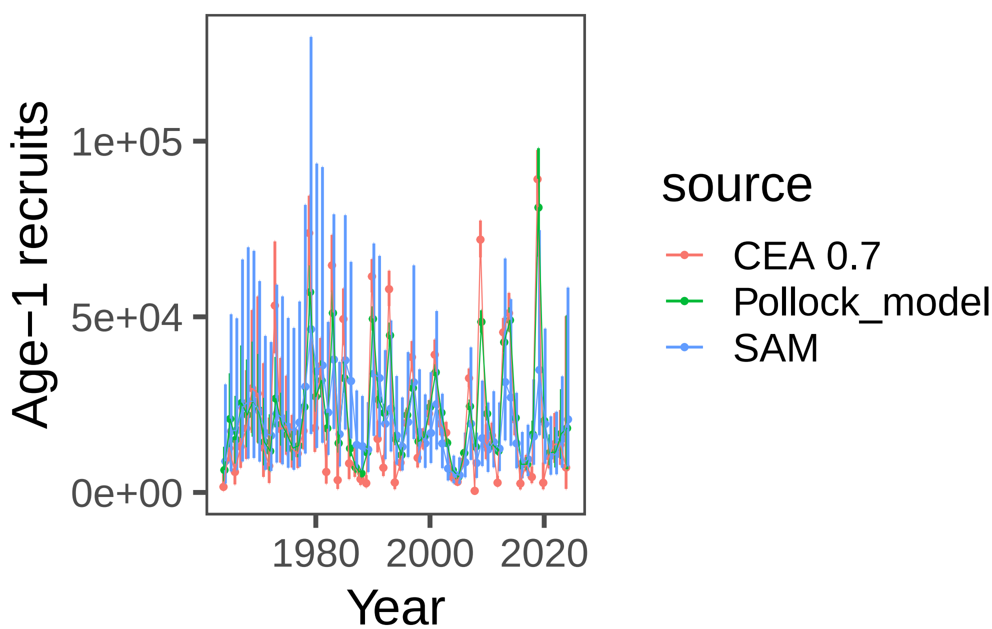

Figure 25: Model results comparing last year's model with the Rceattle model.

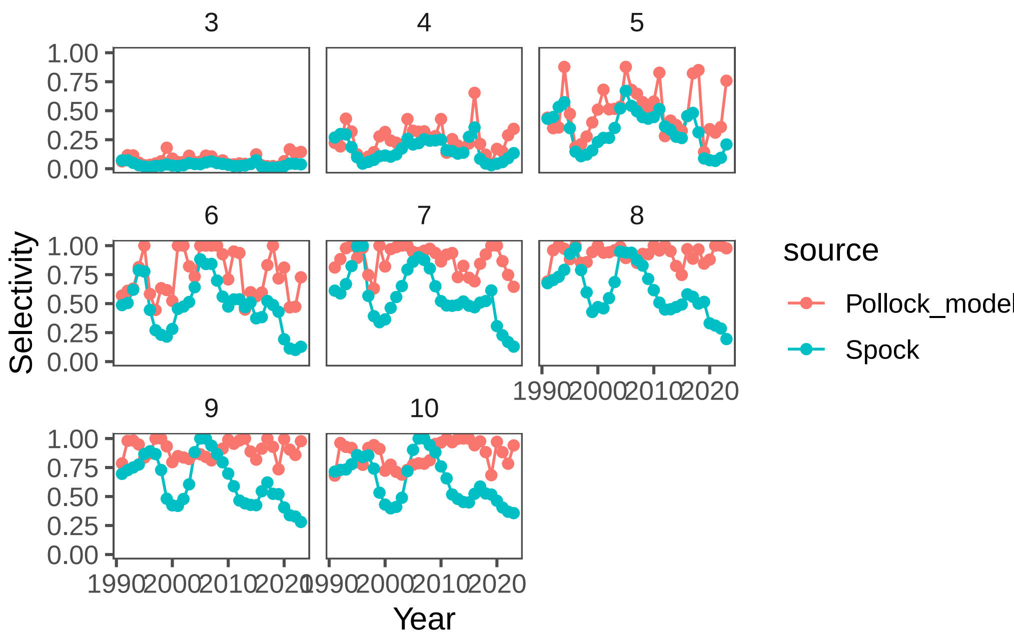

Figure 26: Model results comparing last year's model with the SPock model.

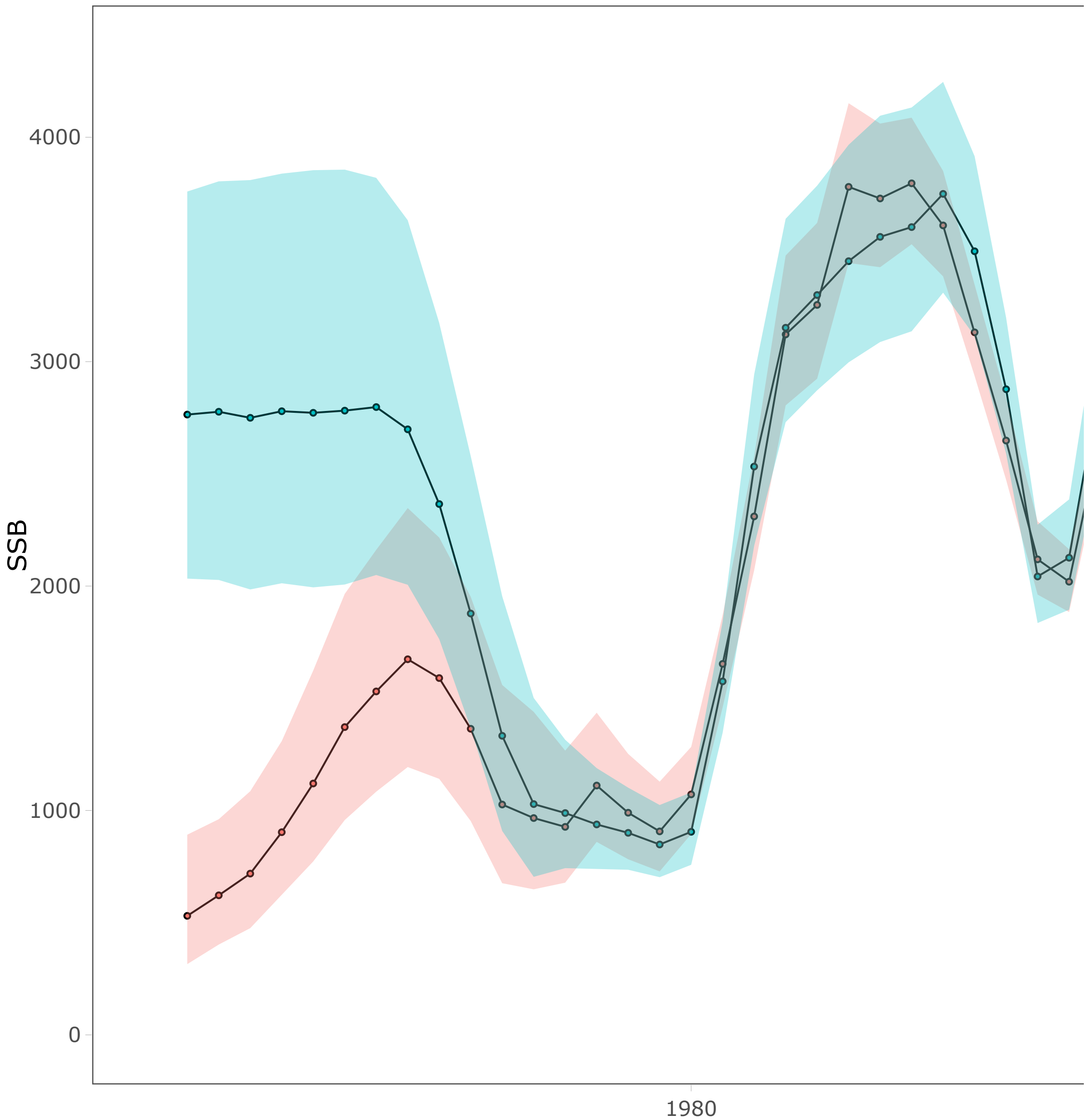

Figure 27: Model results comparing last year's model with the SPock model.

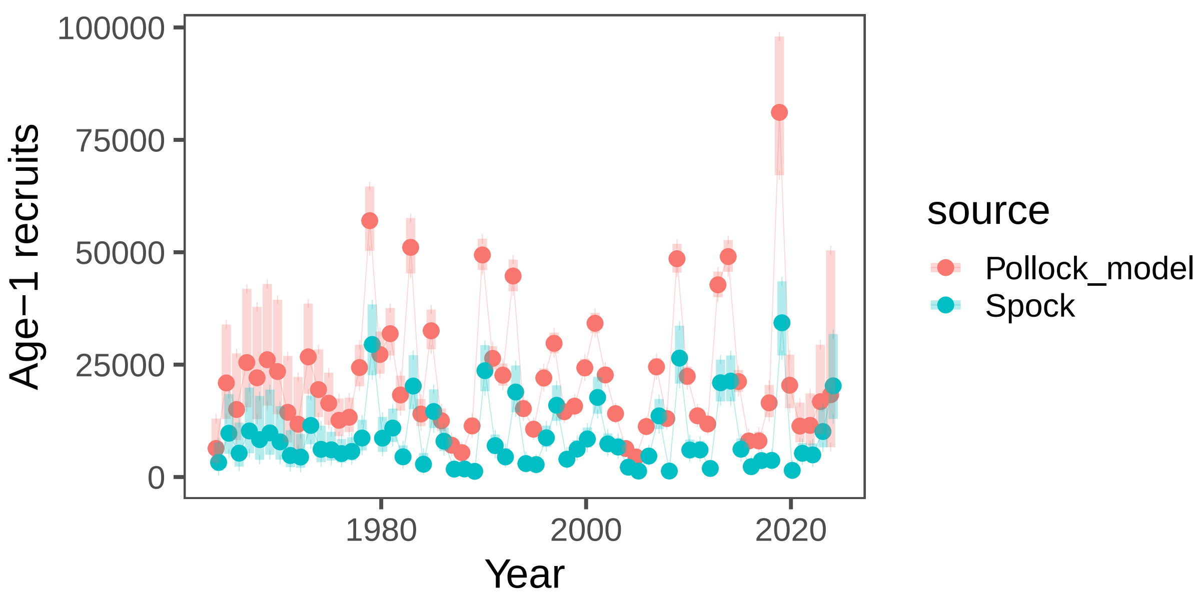

Figure 28: Model results comparing last year's model with the SPock model.

Bottom-trawl survey spatial distribution modeling

Abundance Index Methods

For model-based indices in the Bering Sea, we fitted observations of numerical abundance or biomass per unit area (where the choice of abundance or biomass varied by stock at the request of assessors) from all sampling locations in the 83-112 bottom trawl survey of the EBS, 1982-2025, including:

- Exploratory northern extension samples in 2001, 2005, and 2006
- 83-112 samples available in the NBS in 1982, 1985, 1988, 1991, 2010, 2017-2019, 2021-2023, and 2025

NBS samples collected prior to 2010 and in 2018 did not follow the 20 nautical mile sampling grid that was used in other survey years. Assimilating these unbalanced data requires extrapolating into unsampled areas, facilitated by including a spatially varying response to cold-pool extent (Thorson 2019). This spatially varying coefficient term was estimated for both linear predictors of a delta-model, and detailed comparison of results for EBS pollock has shown that it has a small but notable effect on these indices and resulting stock assessment outputs (O'Leary et al. 2020).

Rather than using the cold-pool covariate for yellowfin sole, we instead used the mean bottom temperature within the inner and middle domain strata from an interpolated temperature product. All models were fitted in the sdmTMB R package (https://pbs-assess.github.io/sdmTMB; Anderson et al. (2022)). All environmental data used as covariates were computed within the coldpool R package (https://github.com/afsc-gap-products/coldpool; Rohan, Barnett, and Charriere (2022)).

We used a spatiotemporal Poisson-link delta-model (Thorson 2018) involving two linear predictors and a gamma distribution to model positive catch rates. We predicted population density across the entire EBS and NBS in each year, using AFSC GAP-approved prediction grids. Spatial and spatiotemporal fields were estimated using a spatial mesh including 250 "knots" whose locations were determined using a K-means algorithm to place the knots evenly over space, in proportion to the dimensions of the prediction grid.

While the prior VAST model was fitted with a 750 knot mesh, we found that sdmTMB models fitted with mesh resolutions from 250-750 knots all provided extremely similar index estimates and that reducing the number of knots dramatically improved computation time. We estimated geometric anisotropy (how spatial autocorrelation declines with differing rates over distance in some cardinal directions than others) and included a spatial and spatiotemporal term for both linear predictors.

To facilitate prediction of density in regions with unsampled years, we specified that the spatiotemporal fields were structured over time as an AR(1) process (where the magnitude of autocorrelation was estimated as a fixed effect for each linear predictor). However, in cases where the AR(1) correlation parameter was estimated to be close to 1 for either model component, the model would be collapsed into a simpler structure by specifying = 1, i.e., modeling spatiotemporal variation as a random walk. We did not include any temporal correlation for intercepts, which we treated as fixed effects for each linear predictor and year. Finally, we used epsilon bias-correction to correct for retransformation bias (Thorson and Kristensen 2016).

We checked model fits for convergence by confirming that the derivative of the marginal likelihood with respect to each fixed effect was sufficiently small (less than ~ 0.001) and that the Hessian matrix was positive definite, along with other model checks produced by the function `sdmTMB::sanity()`. We then checked goodness of model fit by computing Dunn-Smyth randomized quantile residuals (Dunn and Smyth 1996) and visualizing these using a quantile-quantile plot within the DHARMa R package (Hartig 2021). We also evaluated the distribution of these residuals over space in each year, and inspected them for evidence of residual spatiotemporal patterns.

Model-based Age Composition Methods

For model-based estimation of age compositions in the Bering Sea, we fitted observations of numerical abundance-at-age at each sampling location. This was made possible by applying a year-specific, region-specific (EBS and NBS) age-length key to records of numerical abundance and length composition. In subcategories (combinations of year, length, age, sex) that contained insufficient data, age composition was computed from length composition given a globally pooled age-length key.

We computed these estimates in the tinyVAST R package (https://vast-lib.github.io/tinyVAST; Thorson et al. (2024)), assuming a Poisson-link delta-model (Thorson 2018) involving two linear predictors, and a gamma distribution to model positive catch rates. We did not include any density covariates in estimation of age composition for consistency with models used in the previous assessments and due to computational limitations.

We estimated the spatial range assuming geometric isotropy (i.e, spatial autocorrelation declines with equal rates over distance in all cardinal directions). We used the same extrapolation grid as implemented for abundance indices, but here we modeled spatial and spatiotemporal fields with a mesh with a coarser spatial resolution than the index model, here using 50 "knots". This reduction in the spatial resolution of the model, relative to that used for the abundance indices, was necessary due to the increased computational load of fitting multiple age categories.

The estimates of abundance at age were bias-corrected with the same method as the abundance index. We implemented the same diagnostics to check convergence and model fit as those used for the abundance index. Results compared favorably among methods (Figure 29).

Both the model-based bottom trawl index and age composition estimates were calculated using code in the AFSC GAP model-based indices GitHub repository (https://github.com/afsc-gap-products/model-based-indices). Design-based estimates of bottom trawl products were calculated using code in the gapindex R package (https://github.com/afsc-gap-products/gapindex).

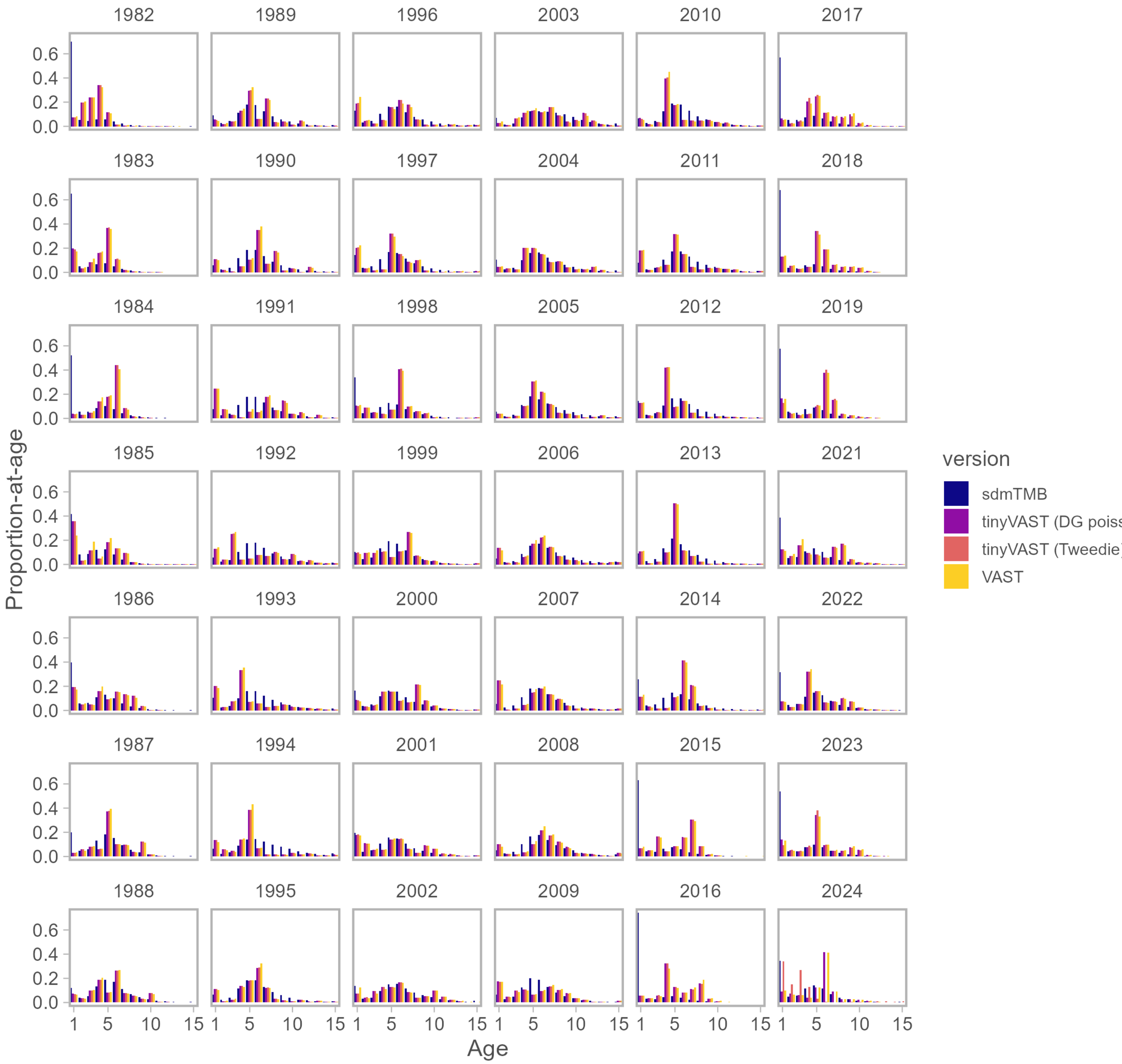

Figure 29: Model results comparing last year's model with the sdmTMB model.

References

Adams, Grant D., Kirstin K. Holsman, Steven J. Barbeaux, Martin W. Dorn, James N. Ianelli, Ingrid Spies, Ian J. Stewart, and André E. Punt. 2022. "An Ensemble Approach to Understand Predation Mortality for Groundfish in the Gulf of Alaska." *Fisheries Research* 251: 106303. https://doi.org/https://doi.org/10.1016/j.fishres.2022.106303.

Anderson, S. C., E. J. Ward, P. A. English, and L. A. K. Barnett. 2022. "sdmTMB: An r Package for Fast, Flexible, and User-Friendly Generalized Linear Mixed Effects Models with Spatial and Spatiotemporal Random Fields." *BioRxiv*, 2022–03. https://doi.org/10.1101/2022.03.24.485545.

Cheng, Matthew. 2025. *SPoCK: Spatial (s) Population (Po) Model with Connectivity (c) Across Generalized Komponents (k).*

Cheng, Matthew L. H., Daniel R. Goethel, Peter-John F. Hulson, Michael J. Wilberg, Craig Marsh, and Curry J. Cunningham. 2025. "Misspecifying Sex-Structured Dynamics in Stock Assessment Models." *Fish and Fisheries*, February, 1–19. https://doi.org/10.1111/faf.12891.

Dunn, K. P., and G. K. Smyth. 1996. "Randomized Quantile Residuals." *Journal of Computational and Graphical Statistics* 5 (1): 1–10. https://doi.org/10.1080/10618600.1996.10474708.

Grandin, C. J., K. F. Johnson, A. M. Edwards, and A. M. Berger. 2024. "Status of the Pacific Hake (Whiting) Stock in U.S. And Canadian Waters in 2024." Joint Technical Committee of the U.S.; Canada Pacific Hake/Whiting Agreement, National Marine Fisheries Service; Fisheries; Oceans Canada.

Hartig, Florian. 2021. *DHARMa: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models*. http://florianhartig.github.io/DHARMa/.

Helser, Thomas E., Irina Benson, Jason Erickson, Jordan Healy, Craig Kastelle, and Jonathan A. Short. 2019. "A Transformative Approach to Ageing Fish Otoliths Using Fourier Transform Near Infrared Spectroscopy: A Case Study of Eastern Bering Sea Walleye Pollock (*Gadus Chalcogrammus*)." *Canadian Journal of Fisheries and Aquatic Sciences* 76 (5): 780–89. https://doi.org/10.1139/cjfas-2018-0112.

Holsman, K. K., J. Ianelli, K. Shotwell, S. Barbeaux, K. Aydin, G. Adams, K. Kearney, A. Sulc, and S. Wassermann. 2024. "Climate-Enhanced Multispecies Stock Assessment for Walleye Pollock, Pacific Cod, and Arrowtooth Flounder in the Eastern Bering Sea." Technical Report. Anchorage, AK: North Pacific Fishery Management Council. https://www.npfmc.org/wp-content/PDFdocuments/SAFE/2024/EBSmultispp.pdf.

Ianelli, Jim. 2025. *Ebswp: Facilitates Model Running for EBS Pollock*. http://afsc-assessments.github.io/EBSpollock/.

Jurado-Molina, J., P. A. Livingston, and J. N. Ianelli. 2005. "Incorporating Predation Interactions to a Statistical Catch-at-Age Model for a Predator-Prey System in the Eastern Bering Sea." *Canadian Journal of Fisheries and Aquatic Sciences* 62 (8): 1865–73.

Methot, Richard D., and Chantell R. Wetzel. 2013. "Stock Synthesis: A Biological and Statistical Framework for Fish Stock Assessment and Fishery Management." *Fisheries Research* 142 (May): 86–99. https://doi.org/10.1016/j.fishres.2012.10.012.

Nielsen, Anders, and Casper W. Berg. 2014. "Estimation of Time-Varying Selectivity in Stock Assessments Using State-Space Models." *Fisheries Research* 158: 96–101. https://doi.org/10.1016/j.fishres.2014.01.014.

Nielsen, Anders, Niels T Hintzen, Henrik Mosegaard, Vanessa Trijoulet, and Casper W Berg. 2021. "Multi-Fleet State-Space Assessment Model Strengthens Confidence in Single-Fleet SAM and Provides Fleet-Specific Forecast Options." *ICES Journal of Marine Science* 78 (6): 2043–52. https://doi.org/10.1093/icesjms/fsab078.

O'Leary, C. A., J. T. Thorson, J. N. Ianelli, and S. Kotwicki. 2020. "Adapting to Climate-driven Distribution Shifts Using Model-based Indices and Age Composition from Multiple Surveys in the Walleye Pollock (Gadus Chalcogrammus) Stock Assessment." *Fisheries Oceanography* 29 (6): 541–57. https://doi.org/10.1111/fog.12494.

Punt, A. E., D. C. Smith, K. KrusicGolub, and S. Robertson. 2008. "Quantifying Age-Reading Error for Use in Fisheries Stock Assessments, with Application to Species in Australia's Southern and Eastern Scalefish and Shark Fishery." *Can. J. Fish. Aquat. Sci.* 65: 1991–2005.

Rohan, S. K., L. A. K. Barnett, and N. Charriere. 2022. "Evaluating Approaches to Estimating Mean Temperatures and Cold Pool Area from AFSC Bottom Trawl Surveys of the Eastern Bering Sea." Tech. Mem. NMFS-AFSC-456. U.S. Dep. Commer., NOAA. https://doi.org/10.25923/1wwh-q418.

Stock, Brian C., and Timothy J. Miller. 2021. "The Woods Hole Assessment Model (WHAM): A General State-Space Assessment Framework That Incorporates Time- and Age-Varying Processes via Random Effects and Links to Environmental Covariates." *Fisheries Research* 240 (August): 105967. https://doi.org/10.1016/j.fishres.2021.105967.

Thorson, J. T. 2018. "Three Problems with the Conventional Delta-Model for Biomass Sampling Data, and a Computationally Efficient Alternative." *Canadian Journal of Fisheries and Aquatic Sciences* 75 (9): 1369–82. https://doi.org/10.1139/cjfas-2017-0266.

———. 2019. "Measuring the Impact of Oceanographic Indices on Species Distribution Shifts: The Spatially Varying Effect of Cold-pool Extent in the Eastern Bering Sea." *Limnology and Oceanography* 64 (6): 2632–45. https://doi.org/10.1002/lno.11238.

Thorson, J. T., S. C. Anderson, P. Goddard, and C. N. Rooper. 2024. "tinyVAST: R Package with an Expressive Interface to Specify Lagged and Simultaneous Effects in Multivariate Spatio-Temporal Models." *arXiv Preprint* arXiv:2401.10193.

Thorson, J. T., and K. Kristensen. 2016. "Implementing a Generic Method for Bias Correction in Statistical Models Using Random Effects, with Spatial and Population Dynamics Examples." *Fisheries Research* 175: 66–74. https://doi.org/10.1016/j.fishres.2015.11.016.